



## AL FLASHER 12 V ENDURANCE TEST REPORT

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|                          |                 |                    |                        |
|--------------------------|-----------------|--------------------|------------------------|
| <b>Part Name</b>         | AL FLASHER 12 V | <b>Application</b> | COMMERCIAL APPLICATION |
| <b>Customer Part No.</b> | FGG00600        | <b>Customer</b>    | ASHOK LEYLAND          |
| <b>UCAL Part No.</b>     | RDEPL/AL/15     | <b>Report No</b>   | R&D/AL/12VF/DUR/26/019 |

### INTRODUCTION:

UCAL is currently supplying 12V AL Flashers to Ashok Leyland for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions.

### PURPOSE:

To evaluate the 12 V flashers assembly after **Endurance Test**.

### TEST SPECIFICATION:

Test standard as per **ISO 16750-2:2023**. DUT should operate within specified functional parameters.

### ACCEPTANCE CRITERIA:

Class A: All function of the flasher performs as designed during and after the test.

### RESULT:

| SAMPLE NO | PERFORMANCE | APPEARANCE | JUDGEMENT |
|-----------|-------------|------------|-----------|
| S:03      | O           | O          | O         |

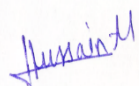
O: OK

X: NG

Δ: To be confirmed

### CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements.

|                                                                                                                  |                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <br><b>Mr. Rahul Bongarde</b> | <br><b>Mr. Hussain Merchant</b> |
| <b>Prepared By</b>                                                                                               | <b>Approved By</b>                                                                                                   |



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### PERFROMACE TEST SPECIFICATIONS:

| SNO | PARAMETERS                             | SPECIFICATIONS                                                   |
|-----|----------------------------------------|------------------------------------------------------------------|
| 1   | Operating voltage                      | 8 V~16 V                                                         |
| 2   | Test voltage                           | 12 V                                                             |
| 3   | Flash rate(Turn signal/Hazard warning) | 60~120 Flashes/Minute                                            |
| 4   | One lamp failure flashing rate         | >140 Flashes/Min(or)1.8 times of flash rate whichever is greater |
| 5   | Duty Cycle                             | 40% ~60%                                                         |

### TEST PROCEDURE:

Standard test circuit with a test voltage of 13.5 V shall be applied for flasher with operation.

Temp: Ambient.

Hazard: Continuous operation with rated load for 36hours

Direction Indicator: Continuous operation with rated load for 100hours.

Temp:  $80 \pm 2^{\circ}\text{C}$

Hazard: Continuous operation with rated load for 5hours

Direction Indicator: Continuous operation with rated load for 30hours.

Temp:  $-10 \pm 2^{\circ}\text{C}$

Hazard: Continuous operation with rated load for 5hours

Direction Indicator: Continuous operation with rated load for 30hours.

Total Duration:206hrs.

### TEST SETUP:





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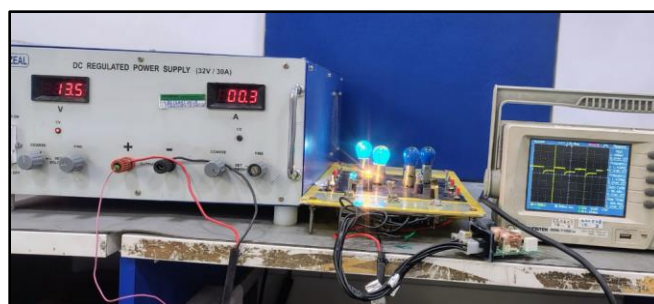
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### TURN SIGNALS & HAZARD SIGNALS:

| LEFT SIGNAL                                                                       | RIGHT SIGNAL                                                                      | HAZARD SIGNAL                                                                       |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |

### WAVEFORM DATA:



### TEST OBSERVATION:

#### Test Observation with Nexperia Part:

| Description                                 | Load                                                   | Observations(at 12 V) | Remark |
|---------------------------------------------|--------------------------------------------------------|-----------------------|--------|
|                                             | LED& CLUSTER                                           | Sample No:01          |        |
| Turn Signal RH                              | 2 x 21 W Lamp + 1x 2W LED+<br>2 W Cluster Signal       | 85                    | OK     |
| Turn Signal LH                              | 2 x 21 W Lamp + 1x 2 W LED+<br>2 W Cluster Signal      | 85                    | OK     |
| Hazard Mode                                 | 2x (2 x 21 W Lamp + 1x 2 W<br>LED+ 2 W Cluster Signal) | 85                    | OK     |
| Fault condition Any one 21W<br>Lamp failure | 2 x 21 W Lamp + 1x 2 W LED+<br>2 W Cluster Signal      | 170                   | OK     |
| Duty cycles for RH&LH                       | 2 x 21 W Lamp + 1x 2 W LED+<br>2 W Cluster Signal      | 50.49%                | OK     |
| Duty cycles for Hazard Mode                 | 2x (2 x 21 W Lamp + 1x 2 W<br>LED+ 2 W Cluster Signal) | 52%                   | OK     |

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.



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### Test Observation with Panjit Part:

| Description                              | Load                                                     | Observations(at 12 V) |            | Remark |
|------------------------------------------|----------------------------------------------------------|-----------------------|------------|--------|
|                                          | LED& CLUSTER                                             | Sample No:03          |            |        |
|                                          |                                                          | Before Test           | After Test |        |
| Turn Signal RH                           | 3x 21 W Lamp + 2 W LED+ 2 W Lamp Cluster Signal          | 85                    | 85         | OK     |
| Turn Signal LH                           | 3x 21 W Lamp + 2 W LED+ 2 W Lamp Cluster Signal          | 85                    | 85         | OK     |
| Hazard Mode                              | 2x (3x 21 W Lamp + 2 W LED)<br>+ 2 W Lamp Cluster Signal | 87                    | 87         | OK     |
| Fault condition Any one 21W Lamp failure | 3x 21 W Lamp + 2 W LED+ 2 W Lamp Cluster Signal          | 172                   | 172        | OK     |
| Duty cycles for RH&LH                    | 3x 21 W Lamp + 2 W LED+ 2 W Lamp Cluster Signal          | 49%                   | 49%        | OK     |
| Duty cycles for Hazard Mode              | 2x (3x 21 W Lamp + 2 W LED)<br>+ 2 W Lamp Cluster Signal | 49%                   | 49%        | OK     |

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.

### TESTED SAMPLES PHOTOS:

| BEFORE TEST SAMPLE – S:03                                                           | AFTER TEST SAMPLE – S:03                                                             |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |

### OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed during & after the test (Class A).